

CENG381 Stochastic Processes

Gambler's Ruin — Practice 1 (Answer Key)

Q1 Solution

a) Model and recurrence:

States: current wealth of Alice, $\{0, 1, 2, \dots, 8\}$. State 0: Alice bankrupt (absorbing, $p_k=0$). State 8: Bob bankrupt (absorbing, $p_k=1$).

$$\begin{aligned}\text{Recurrence: } p_k &= (1/2) \cdot p(k+1) + (1/2) \cdot p(k-1) \quad \text{for } 1 \leq k \leq 7 \\ \text{Boundary: } p_0 &= 0, \quad p_8 = 1\end{aligned}$$

b) Solving the recurrence:

Re-write as $p(k+1) - p_k = p_k - p(k-1)$: the differences are constant. So p_k is linear in k : $p_k = \alpha + \beta \cdot k$.

$$\begin{aligned}p_0 = 0 &\rightarrow \alpha = 0 \\ p_8 = 1 &\rightarrow \beta \cdot 8 = 1 \rightarrow \beta = 1/8 \\ p_k &= k/8 \quad \text{for } k = 0, 1, \dots, 8 \\ p_3 &= 3/8 = 0.375\end{aligned}$$

c) Loss probability and effect of doubling stake:

$$P(\text{Alice loses} \mid \text{start } k=3) = 1 - p_3 = 1 - 3/8 = 5/8$$

$$\begin{aligned}\text{If Alice starts with } k=6, \text{ Bob has } k'=2: \\ p_6 &= 6/8 = 3/4 = 0.75\end{aligned}$$

Doubling her stake from \$3 to \$6 raises Alice's win probability from 37.5% to 75% — the win probability scales linearly with the initial stake fraction k/n .

Q2 Solution

a) Recurrence and boundary conditions:

$$\begin{aligned}f_k &= 1 + (1/2) \cdot f(k+1) + (1/2) \cdot f(k-1) \quad \text{for } 1 \leq k \leq n-1 \\ f_0 &= 0, \quad f_n = 0\end{aligned}$$

b) Verification that $f_k = k(n-k)$ solves the recurrence:

$$\begin{aligned}f(k+1) &= (k+1)(n-k-1) = kn - k^2 - 2k + n - 1 \\ f(k-1) &= (k-1)(n-k+1) = kn - k^2 + 2k - n - 1\end{aligned}$$

CENG381 Stochastic Processes

Gambler's Ruin — Practice 1 (Answer Key)

$$f(k+1) + f(k-1) = 2kn - 2k^2 - 2$$

$$\begin{aligned} 1 + (1/2) \cdot [f(k+1) + f(k-1)] \\ &= 1 + (1/2) \cdot (2kn - 2k^2 - 2) \\ &= 1 + kn - k^2 - 1 \\ &= kn - k^2 \\ &= k(n-k) = f_k \quad \checkmark \end{aligned}$$

$$\text{Boundary: } f_0 = 0 \cdot (n-0) = 0 \quad \checkmark \quad f_n = n \cdot (n-n) = 0 \quad \checkmark$$

c) Values for $n=8$, $k=3$ and maximum:

$$f_3 = 3 \cdot (8-3) = 3 \cdot 5 = 15 \text{ rounds}$$

$$f_k = k(n-k) \text{ is maximised at } k^* = n/2 = 4:$$

$$f_4 = 4 \cdot (8-4) = 4 \cdot 4 = 16 \text{ rounds (maximum)}$$

The game lasts longest when the two players start with equal stakes. From $k=3$, the expected game length is 15 rounds.

Q3 Solution

a) Biased-walk recurrence and general solution:

$$\text{Recurrence: } p_k = (2/3) \cdot p(k+1) + (1/3) \cdot p(k-1) \quad \text{for } 1 \leq k \leq 4$$

$$\text{Rearranged: } (2/3) \cdot p(k+1) - p_k + (1/3) \cdot p(k-1) = 0$$

$$\text{Characteristic equation: } (2/3) \cdot r^2 - r + 1/3 = 0$$

$$\text{Roots: } r=1 \text{ and } r = (1/3)/(2/3) = 1/2 \quad [\text{i.e. } r = q/p]$$

$$\text{General solution: } p_k = A + B \cdot (1/2)^k$$

b) Applying boundary conditions:

$$p_0 = 0: \quad A + B \cdot 1 = 0 \quad \rightarrow \quad A = -B$$

$$p_5 = 1: \quad -B + B \cdot (1/2)^5 = 1$$

$$B \cdot (1/32 - 1) = 1$$

$$B \cdot (-31/32) = 1$$

CENG381 Stochastic Processes

Gambler's Ruin — Practice 1 (Answer Key)

$$B = -32/31, \quad A = 32/31$$

$$\begin{aligned} p_k &= (32/31) \cdot [1 - (1/2)^k] \\ &= [1 - (1/2)^k] / [1 - (1/2)^5] \end{aligned}$$

c) Compute p_2 and compare:

$$\begin{aligned} p_2 &= (32/31) \cdot [1 - (1/2)^2] \\ &= (32/31) \cdot (3/4) \\ &= 24/31 \approx 0.774 \end{aligned}$$

$$\text{Unbiased: } p_2 = 2/5 = 0.400$$

$$\text{Increase: } 0.774 - 0.400 = 0.374 \quad (+37.4 \text{ percentage points})$$

The rightward drift ($p=2/3$) dramatically improves Alice's chances: from 40% in the fair game to about 77% in the biased game. This reflects that the biased walk spends much more time moving towards the winning boundary.